

CLASS 21+22: LEARNING RESOURCES

Authentication & Build Process · Curated References

Authentication

Build & Env

■ AI Workflow

01 React Context & Authentication

React Docs — Passing Data Deeply with Context

<https://react.dev/learn/passing-data-deeply-with-context>

The official Context guide. Covers createContext, Provider, and useContext. Read this to understand WHY Context solves the prop-drilling problem for auth state.

JWT.io — Debugger & Explainer

<https://jwt.io>

Paste any JWT token and this site decodes it into its three parts (header, payload, signature). Use it in class to inspect the token returned by reqres.in and see what data is inside.

reqres.in — Free Mock Auth API

<https://reqres.in>

Free API for testing auth flows. No signup needed. Supports login and register endpoints. Test credentials: eve.holt@reqres.in / cityslicka.

React Router — Navigate Component

<https://reactrouter.com/en/main/components/navigate>

Official docs for the Navigate component used in ProtectedRoute. Explains the replace prop and when to use Navigate vs useNavigate().

02 Build Process & Environment Variables

Vite Docs — Env Variables and Modes

<https://vitejs.dev/guide/env-and-mode>

Official Vite documentation for environment variables. Covers the VITE_ prefix requirement, import.meta.env, and .env file types. Bookmark this — you will reference it every time you add a new API key.

Vite Docs — Building for Production

<https://vitejs.dev/guide/build>

Covers npm run build, the dist/ output folder, and build optimisations. Read this before your first deployment next class.

React Docs — lazy() and Suspense

<https://react.dev/reference/react/lazy>

Official reference for React.lazy() and Suspense. Shows the exact code-splitting pattern used in class. The fallback prop on Suspense is what shows while the lazy component is loading.

GitHub Docs — Ignoring Files (.gitignore)

<https://docs.github.com/en/get-started/getting-started-with-git/ignoring-files>

Complete .gitignore guide. Shows how to add files that were accidentally committed and how patterns work. Essential reading before your first deployment.

03 YouTube Tutorials

Web Dev Simplified — React Context Tutorial

Search YouTube: "Web Dev Simplified React Context tutorial"

Clear explanation of React Context from creation to consumption. Covers the exact AuthContext pattern from class with a different example.

Traversy Media — JWT Authentication in React

Search YouTube: "Traversy Media JWT auth React"

End-to-end React auth walkthrough: JWT, localStorage, protected routes, and Context. Reinforces everything from class from a different angle.

Fireship — Vite in 100 Seconds

Search YouTube: "Fireship Vite explained 100 seconds"

Quick overview of what Vite does and why the build process is faster than older tools.

04 ■ AI Prompt Templates

Copy and adapt these for your homework. Always read and understand the response first.

AuthContext setup

Generate a complete AuthContext.jsx using function declarations (not arrow functions). Include: user state initialised from localStorage, login(userData) that saves to localStorage, logout() that clears localStorage, and a useAuth() custom hook.

ProtectedRoute

Write a ProtectedRoute component for React Router v6 using a function declaration. It should call useAuth() to check if user is logged in. If not, redirect to /login using <Navigate> with replace=true. If yes, render children.

Debugging auth redirect loop

My ProtectedRoute redirects to /login even when I'm logged in. User is stored in localStorage. Here is my AuthContext and ProtectedRoute: [paste]. What is wrong? Explain the cause before giving the fix.

Move API keys to .env

I have API keys hardcoded in my React components. Help me move them all to a .env file using Vite's VITE_ prefix convention. Also create the .env.example content. Here are my components: [paste]

Convert to lazy loading

Convert my App.jsx to use React.lazy() for all page imports and wrap Routes in a Suspense component. Keep function declarations throughout. Here is my current App.jsx: [paste]

05 Quick Reference — Auth + Build Patterns

```
// AuthContext pattern: const AuthContext = createContext(null) export function AuthProvider({
children }) { const [user, setUser] = useState( () => JSON.parse(localStorage.getItem('user')) )
function login(u) { setUser(u); localStorage.setItem('user', JSON.stringify(u)) } function logout()
{ setUser(null); localStorage.removeItem('user') } return <AuthContext.Provider
value={{user,login,logout}}>{children}</AuthContext.Provider> } export function useAuth() { return
useContext(AuthContext) } // ProtectedRoute pattern: export default function ProtectedRoute({
children }) { const { user } = useAuth() return user ? children : <Navigate to='/login' replace /> }
// .env variable in code: const KEY = import.meta.env.VITE_MY_KEY // Lazy loading: const Page =
lazy(() => import('./pages/Page')) <Suspense
fallback=<p>Loading...</p>><Routes>...</Routes></Suspense>
```